

Drilltek

A Borehole Management Application for Diamond Drilling in the Mining Industry

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Background

Drillcores extracted using Diamond Drill.

Drillcores logged for geological information and assays.

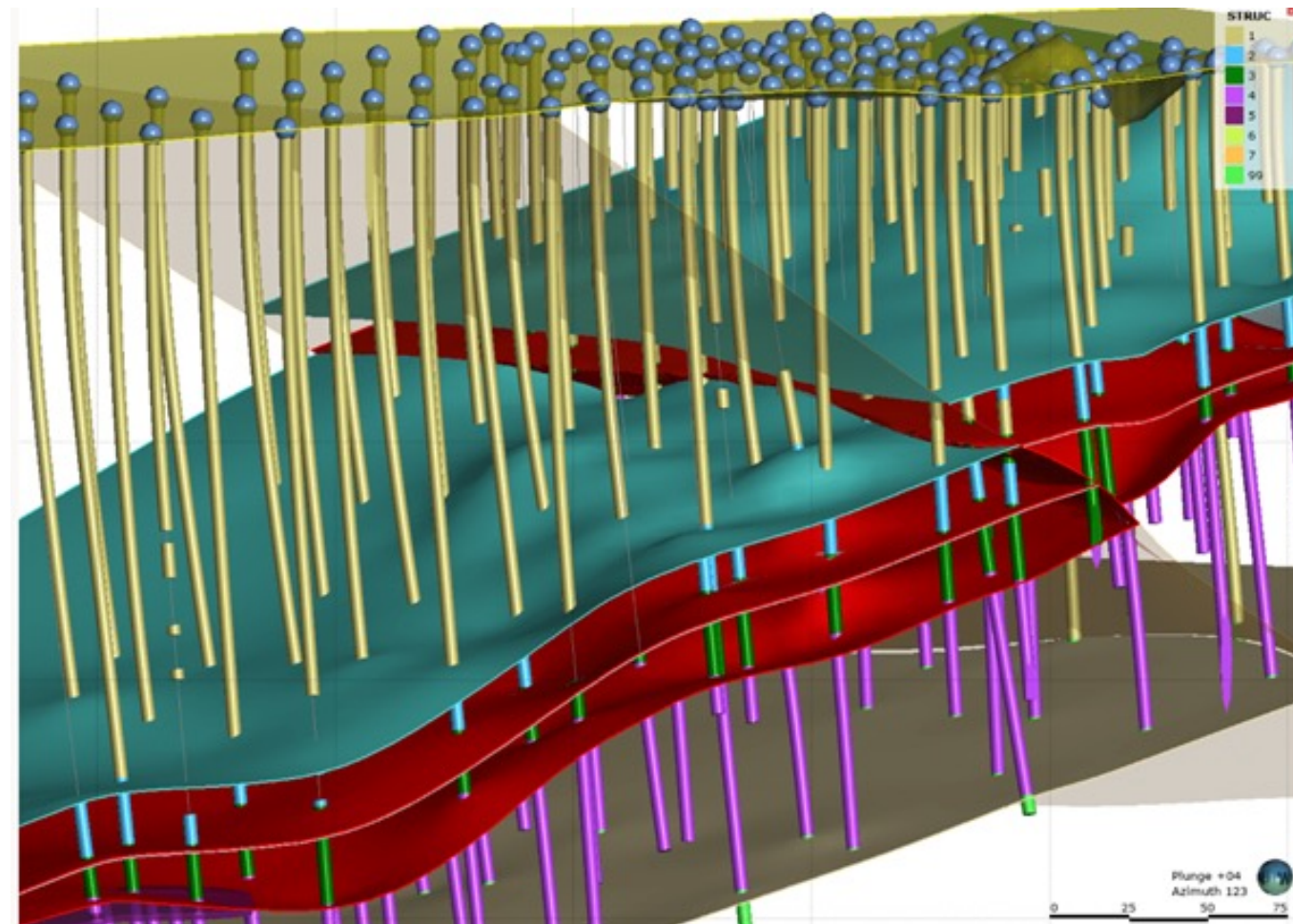
Information stored in a Database.

Stored information used for 3D Modelling.









Motivation



Traditionally done using pen and paper.



Information transcribed and uploaded to Database.



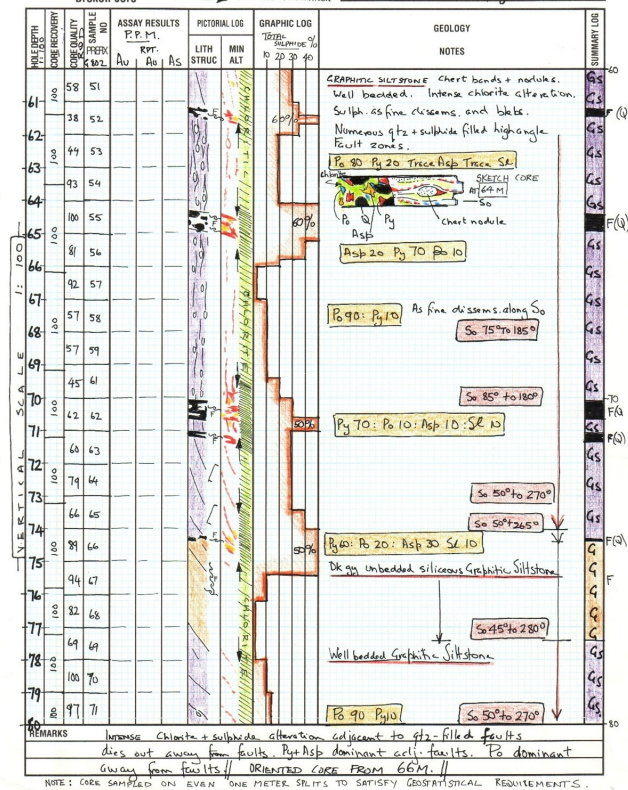
Systems being released carry high license fees or incomplete functionality.

RGC EXPLORATION PTY LTD

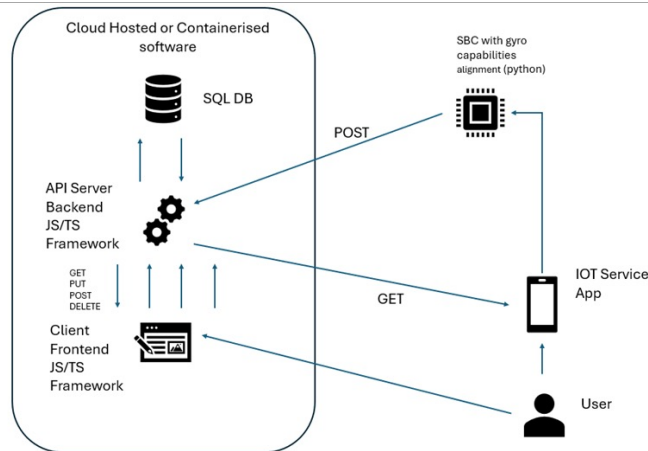
DRILL HOLE No MP DDH 152SHEET 4 OF 5

- Bedding
Cleavage
Foliation
Fault, Shear
Breccia
Broken core
- Disseminated
Massive
Pervasive Alteration
Narrow vein
Visible gold
Increase Alteration

PROJECT : MOUNT PORTER
PROSPECT : 10 400 NORTH
DATE : 11th July 1992
LOGGED BY : R.W. Majori banks



Aim



- ▶ Create web-based system for borehole management and logging.
- ▶ Build using open-source tools.
- ▶ Supported by relational database accessible by modelling software.



► DEMONSTRATION



► PLANNING

Market Research

Analysed repos on Github,
Bitbucket.

Analysed commercial and open-
source products.

Confirmed no open-source offerings.

Identified desirable functions from
findings.

STACK SELECTION



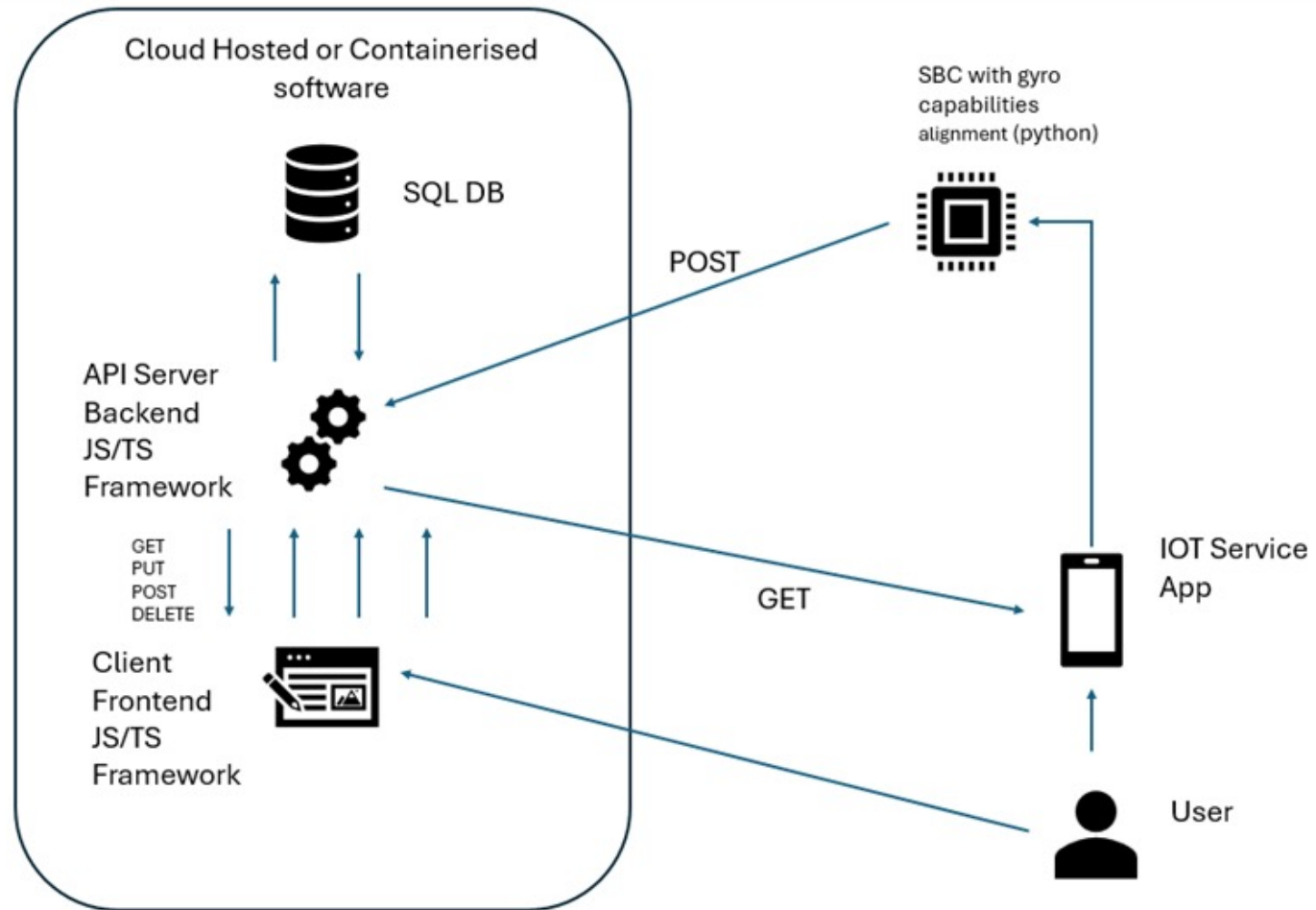
Advised to leverage AI.

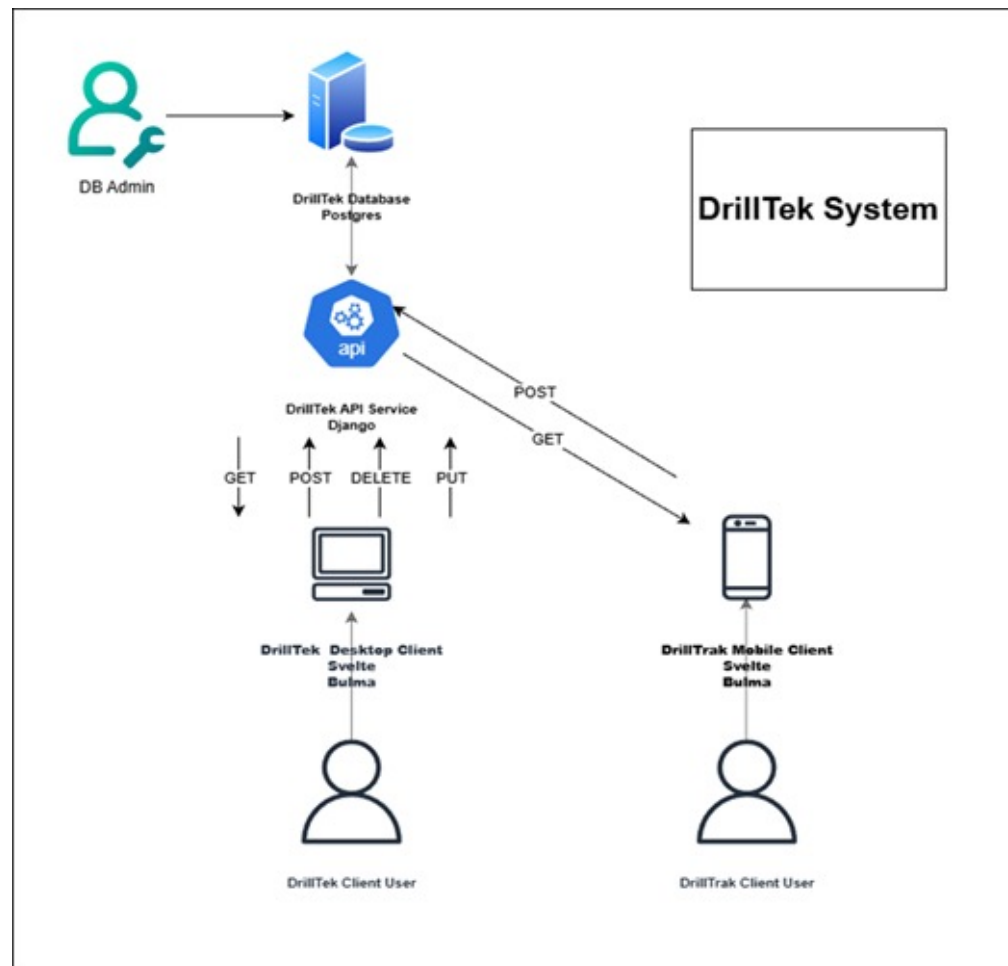


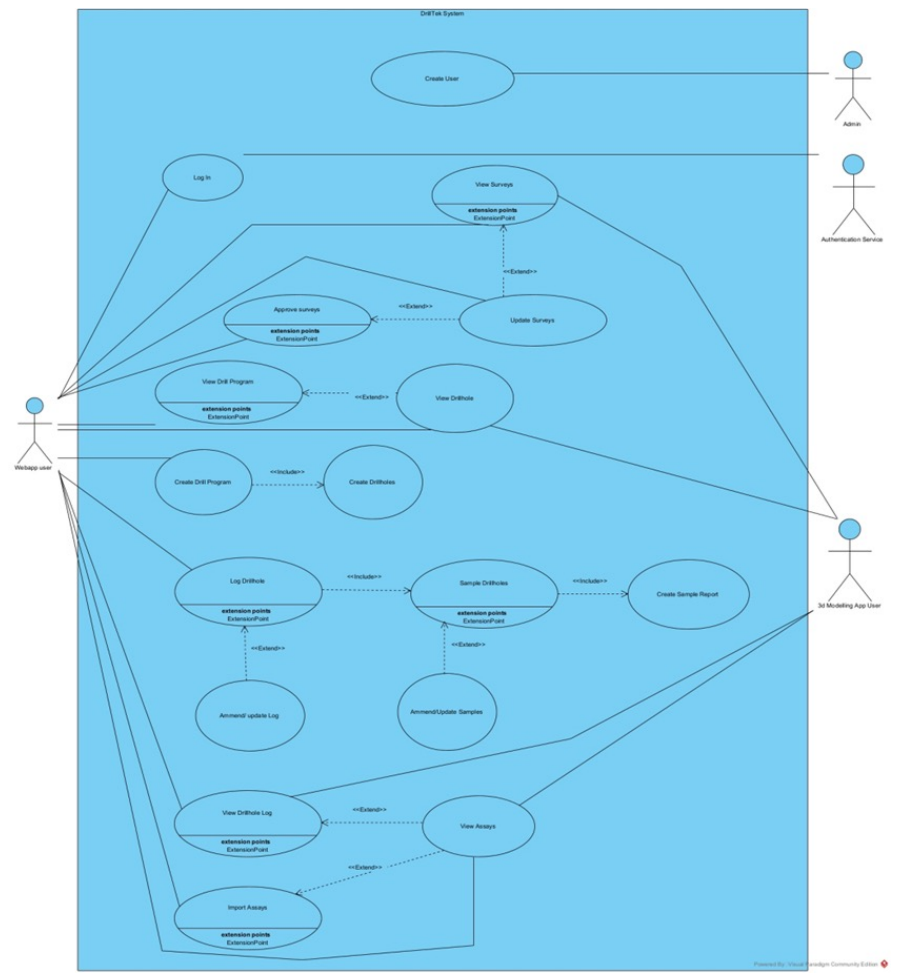
Developed software specification
for input.

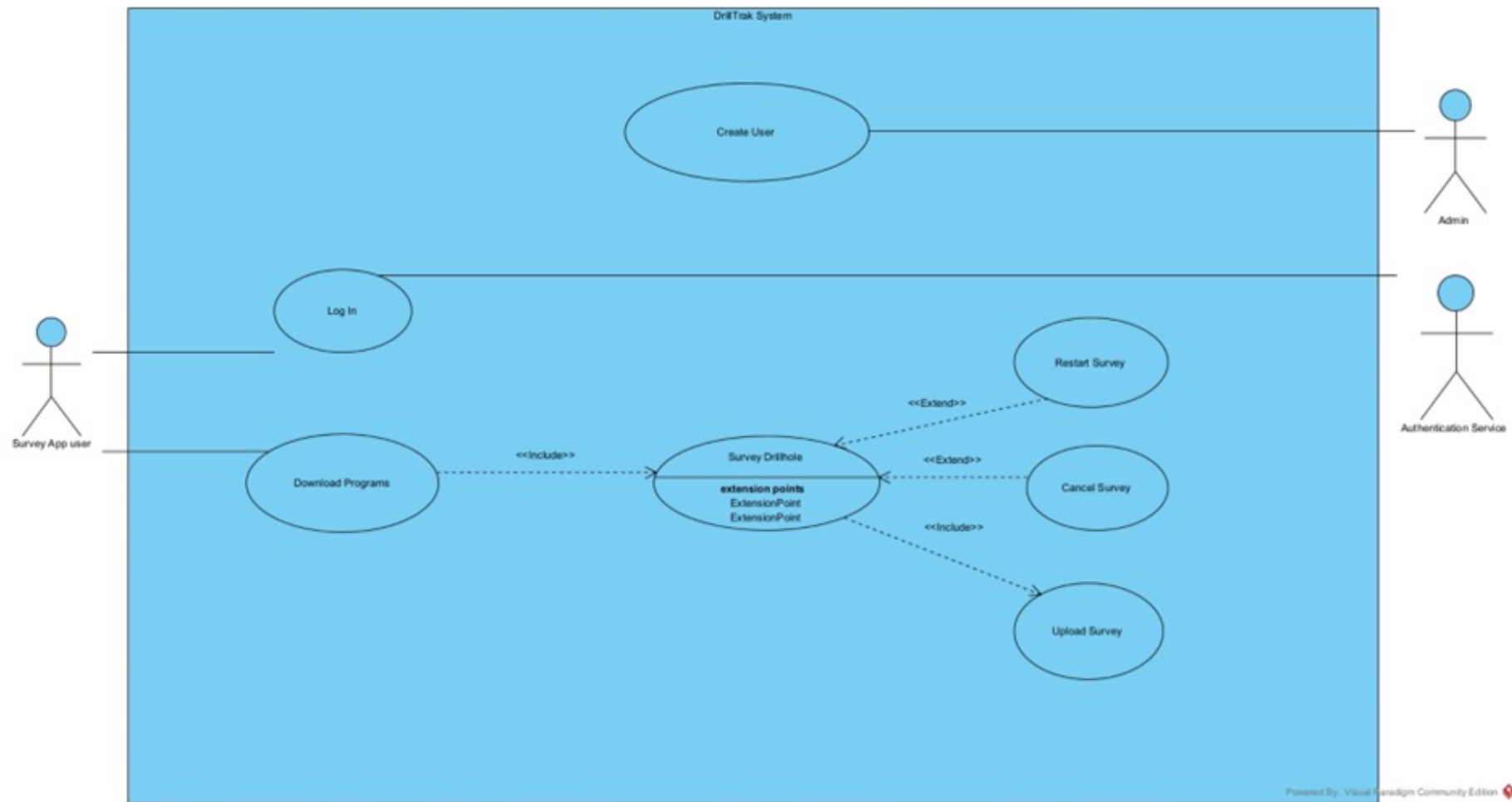


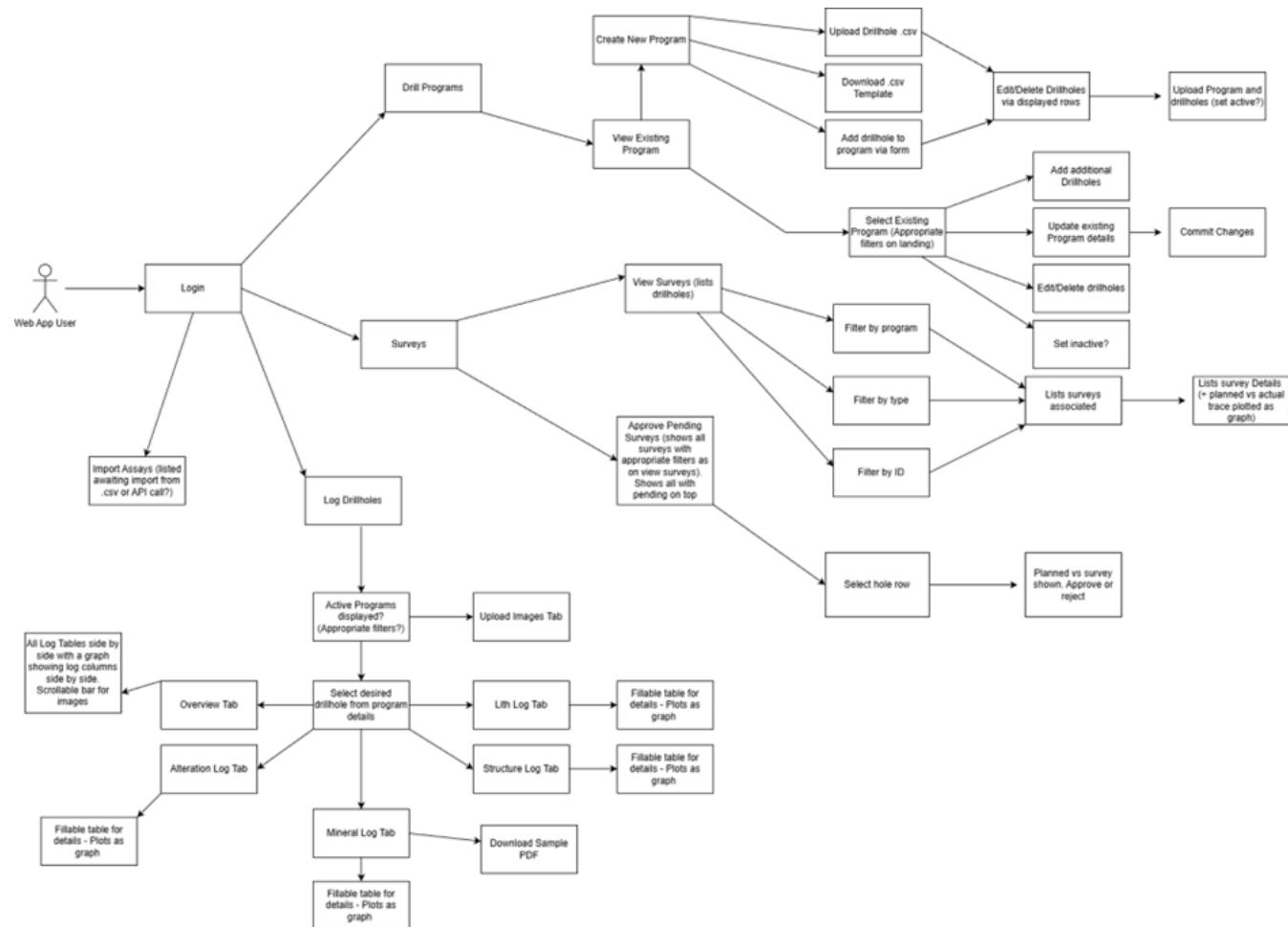
Fed in for feedback based on
appropriateness for architecture
and development timeframe.

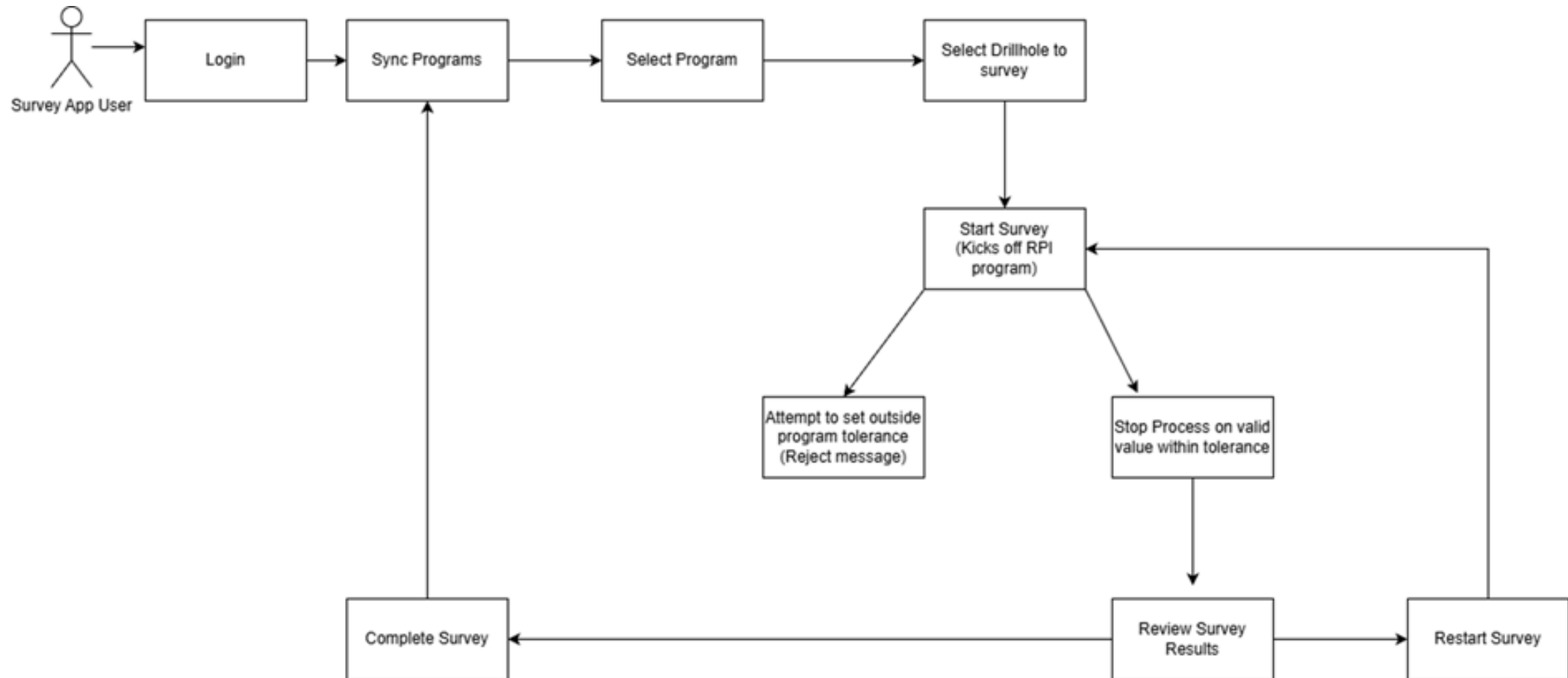












Welcome To DrillTek

All Things Diamond Drilling, All in One
Place

Login



Login

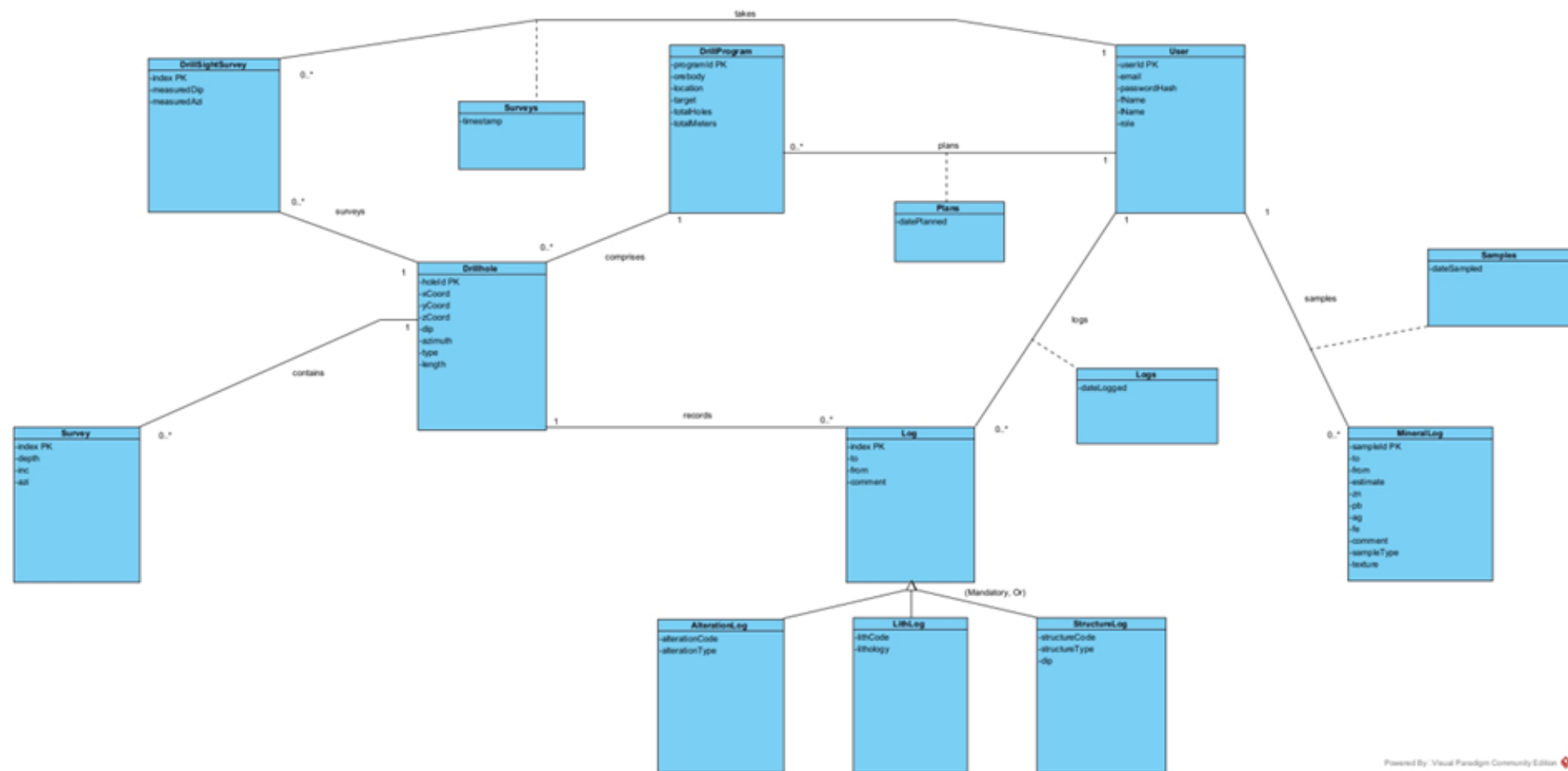
Please Enter Your Credentials to Log In

Email

Password

Login





User(userId,email,passwordHash,fName,lName,role)
Primary Key userId

USER STORIES + PROJECT

► MANAGEMENT DEMO

Views Reqs (1/1) Issue types Parent Label Sprint Assignee Release Estimate Author Milestone

GOALS AND PLANNING

New Issues
No limit
8 Issues / 81 Points

- Feature Drilthole #20
Open drillhole logging
Feature
Sprint Feb 2 - Feb 16, 2026
Drilthole web app
- Feature Drilthole #26
Log summary page
Feature
Sprint Feb 2 - Feb 16, 2026
Drilthole web app
- Feature Drilthole #21
drillhole lithology logging
Feature
Sprint Feb 2 - Feb 16, 2026
Drilthole web app
- Feature Drilthole #22
drillhole structure logging
Feature
Sprint Feb 2 - Feb 16, 2026
Drilthole web app
- Feature Drilthole #23
drillhole alteration logging
Feature
Sprint Feb 2 - Feb 16, 2026
Drilthole web app
- Feature Drilthole #24
drillhole mineral logging
Feature
Sprint Feb 2 - Feb 16, 2026
Drilthole web app

Icebox
No limit
9 Issues / 83 Points

- Feature Drilthole #1
Filter drilling portal
Feature
Drilthole web app
- Feature Drilthole #11
Add drillhole by uploading .csv
Feature
Drilthole web app
- Feature Drilthole #13
Filter drillholes
Feature
Drilthole web app
- Feature Drilthole #12
Download a .csv template for uploading drillholes
Feature
Drilthole web app
- Feature Drilthole #25
Export sample pdf and .csv
Feature
Drilthole web app
- Feature Drilthole #27
Import assays
Feature
Drilthole web app
- Feature Drilthole #17
Filter Logging portal Landing
Feature
Drilthole web app

Backlog
No limit
7 Issues / 27 Points

- Feature Drilthole #5
Drilling portal landing
Feature
Sprint Jan 19 - Feb 2, 2026
Drilthole web app
- Feature Drilthole #7
Create drill program
Feature
Sprint Jan 19 - Feb 2, 2026
Drilthole web app
- Feature Drilthole #8
Open drill program
Feature
Sprint Jan 19 - Feb 2, 2026
Drilthole web app
- Feature Drilthole #6
Edit drill program
Feature
Sprint Jan 19 - Feb 2, 2026
Drilthole web app
- Feature Drilthole #10
Add drillhole using form
Feature
Sprint Jan 19 - Feb 2, 2026
Drilthole web app
- Feature Drilthole #14
Open drillhole
Feature
Sprint Jan 19 - Feb 2, 2026
Drilthole web app
- Feature Drilthole #15

In Progress
No limit
1 Issue / 5 Points

- Feature Drilthole #4
Log in
Feature
Sprint Feb 2 - Feb 16, 2026
Drilthole web app

Review/QA
No limit
0 Issues / 0 Points

Review/QA
Issues open to the team for review and testing. Code complete, pending feedback.
[Create a new issue](#)

Done
No limit
1 Issue / 5 Points

- Task Drilthole #21
Create users as admin
Feature
Sprint Jan 19 - Feb 2, 2026
Drilthole web app

Closed
No limit
-0 Issues / 0 Points



► DEVELOPMENT

Sprint 1

Sprint 3

Sprint 5

2-16 Feb.
2026

2-16 Mar.
2026

19 Jan. - 2
Feb. 2026

16 Feb. - 2
Mar. 2026

16-30 Mar.
2026

Sprint 2

Sprint 4

Outcome 1



SVELTE CLIENT APP
BUILT WITH SSR
PATTERN.



DJANGO API USING
CLASS ACTION PATTERN
IN DRF WITH JWT AUTH.



POSTGRESQL DATABASE.

Outcome 2



APPLICATION DESIGNED TO
BE USED INTERNALLY.



APPLICATION STILL IN
DEVELOPMENT STATE.



DOCKER IMAGES AND
COMPOSE SCRIPT
DEVELOPED TO RUN IN
ORCHESTRATED CONTAINER.

LEARNINGS AND FUTURE

▶ DEVELOPMENT

Learnings

An open source based version of system can be created.

Learned new framework in Python

Got more comfortable with client rendered frontends.

Future works

Several stories not completed.

RBAC to sections of site.

Export sample .csv and PDF.

Import assays.

Page pagination.

An abstract digital cityscape composed of glowing blue cubes and rectangular structures. The surfaces of these structures are covered in a dense pattern of binary code (0s and 1s). Several bright blue and red light beams emanate from the structures, creating a sense of depth and digital activity. The background is a dark blue gradient, and the overall aesthetic is futuristic and technological.

Future works 2

- ▶ Drillsight system planned but not developed .
- ▶ Building on attempt in Comp-Sys module.
- ▶ Svelte mobile client integrated with RPI-5.

References

- ▶ Shutterstock (2026) core-drilling-rig. Available at: <https://www.shutterstock.com/search/core-drilling-rig> (Accessed 28-03-2026)
- ▶ Shutterstock (2026b) Geologist-field. Available at: <https://www.shutterstock.com/search/geologist-field> (Accessed 28-03-2026)
- ▶ Seequent (2026) *Experience-40-70 reductions in drillhole loading with leapfrog*. Available at: <https://www.seequent.com/experience-40-70-reductions-in-drillhole-loading-with-leapfrog/> (Accessed 28-03-2026)
- ▶ rogermarjoribanks (2015) *Diamond-drill-core-log-sheets*. Available at: <http://rogermarjoribanks.info/diamond-drill-core-log-sheets/> (Accessed 28-03-2026)
- ▶ Google Play (2026) Mxdeposit. Available at: <https://play.google.com/store/apps/details?id=net.minalytix.mxdeposit> (Accessed 28-03-2026)